

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry.	High	Pass: Alert queued for certificate expiring in 15 days. Fail: Expired certificate ignored without notification.	Ensures SSL certificates are monitored and alerts are generated before expiry.
FR-002	Functional	The system shall perform a daily WHOIS audit of each monitored domain and retrieve its registration expiry date.	High	Pass: Registration expiry date is retrieved for a valid monitored domain. Fail: Domain expiry information is not retrieved or recorded.	Allows the system to detect upcoming domain registration expiry.
FR-003	Functional	The system shall generate escalation alerts when a monitored domain or SSL certificate approaches its expiry date.	High	Pass: An alert is generated when a configured expiry threshold is reached. Fail: No alert is generated after the threshold is reached.	Ensures responsible personnel are informed before domain or certificate expiry.
FR-004	Functional	The system shall allow the SysAdmin to add, update, and remove domains from the monitoring list.	High	Pass: A valid domain added by the SysAdmin appears in the monitoring list. Fail: An invalid domain is added without validation.	Allows the SysAdmin to maintain the domains being monitored.

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FR-005	Functional	The system shall allow the SysAdmin and Security Officer to view the WHOIS and SSL/TLS audit status of monitored domains.	Medium	Pass: The system displays the domain registration expiry date, certificate expiry date, and current status. Fail: Available expiry information is missing from the status view.	Provides users with a centralized view of domain and certificate status.
NFR-001	Non-Functional — Performance & Security	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load.	Ensures efficient daily monitoring of a large number of domains.
NFR-002	Non-Functional — Reliability	The system shall retry a failed WHOIS or TLS audit before marking the audit as unsuccessful.	High	Pass: A failed audit is retried before being marked unsuccessful. Fail: A temporary audit failure is immediately marked unsuccessful without retrying.	Reduces incorrect failures caused by temporary network or service issues.

Use-Case Flow Specification

Use Case: UC-03 — Perform TLS Certificate Audit

Preconditions

1. At least one valid domain is registered in the monitoring system.
2. The monitored domain has an accessible SSL/TLS endpoint.
3. The daily monitoring process has been initiated.

Main Success Scenario

1. The system starts the scheduled daily TLS certificate audit.
2. The system selects a monitored domain.
3. The system initiates a TLS handshake with the domain.
4. The system retrieves the SSL/TLS certificate.
5. The system extracts the certificate expiration date.
6. The system calculates the remaining time until certificate expiry.
7. The system compares the expiry date against the configured thresholds of 30, 15, and 3 days.
8. If an expiry threshold is reached, the system generates an expiry alert.
9. The system records the audit result and certificate status.
10. The system proceeds to the next monitored domain.
11. The audit ends successfully after all monitored domains have been checked.

Alternate Flow — TLS Handshake Failure

3a. The TLS handshake fails.

3a1. The system detects that the TLS handshake could not be completed.

3a2. The system retries the TLS handshake.

3a3. If the retry succeeds, the system continues from Step 4.

3a4. If the retry fails again, the system records the audit as unsuccessful.

3a5. The system records the failure for follow-up and proceeds to the next monitored domain.

Postconditions

1. The certificate expiry date and audit result are stored when the audit succeeds.
2. The certificate status is updated.
3. An expiry alert is generated when a configured expiry threshold is reached.
4. Failed audits are recorded for follow-up.

